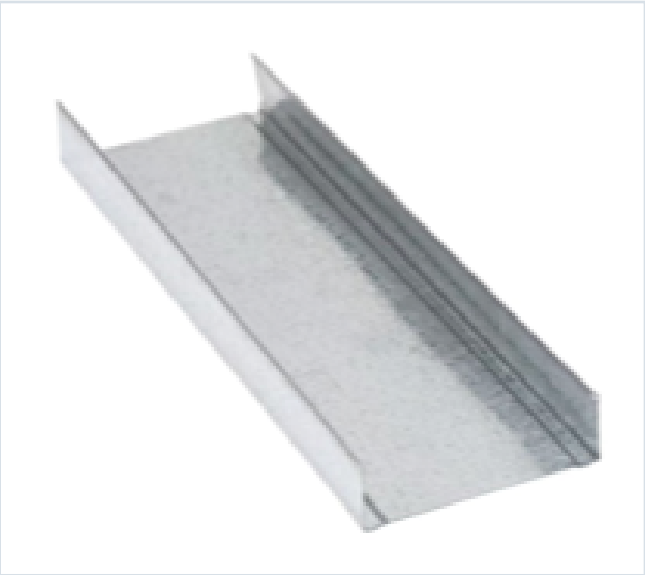


DECKING PRODUCT SUBMITTAL

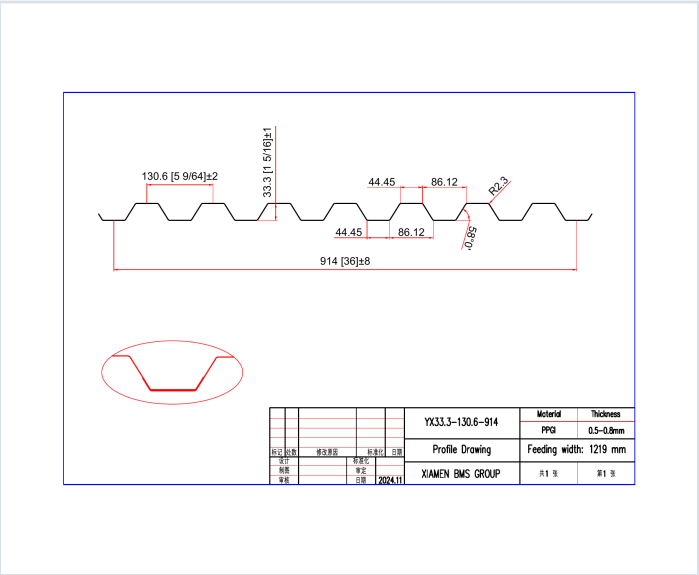
1316FD-FY50-18-G90

PRODUCT	GRADE	GAUGE	COATING
1 5/16" Form Deck	Grade 50	18 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.6	50	0.310	0.294	9291	8812	0.290	0.257

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 70, 71
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

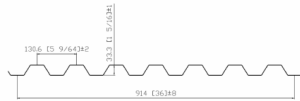
1316FD-FY50-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 70

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	E, ksi	S _x (in ³ /ft)	S _y (in ³ /ft)	ASD (0 + 16%) M _u / 0 (in ³ -lb/ft)	I _x (in ⁴ /ft)	I _y (in ⁴ /ft)
22	0.0096	1.6	50	0.179	0.170	6308	608	0.107
20	0.0108	2.0	50	0.222	0.216	6981	6407	0.210
18	0.0124	2.6	50	0.300	0.294	9208	8962	0.290
16	0.0148	3.0	50	0.390	0.378	11667	11327	0.363

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S100-2017 and AISI S100-2018.

Shear and Web Crippling

Gage	V _u (lb/ft)	Web Crippling (R _u), lb/ft One Flange Loading End Bearing	Web Crippling (R _u), lb/ft One Flange Loading Interior Bearing
22	2424	801	880
20	3053	1043	1135
18	5032	1894	2087
16	6299	2591	2833

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S100-2017 and AISI S100-2018.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	143	138	133	128	123	118	113	108	103
	20	178	147	125	105	91	79	69	59	49
	18	248	226	172	147	126	103	89	76	63
	16	311	257	216	184	159	138	122	108	96
Double	22	158	152	146	140	134	128	122	116	110
	20	172	142	120	102	88	77	67	60	53
	18	235	194	163	139	120	104	92	81	73
	16	292	230	190	164	144	128	115	103	94
Triple	22	170	163	156	150	143	136	130	123	116
	20	275	178	149	127	110	96	84	74	66
	18	284	243	204	174	150	131	115	102	91
	16	378	302	262	223	193	168	147	131	117

- Notes:
- All section properties and ASD (0 + 16%) uniform loads are calculated in accordance with AISI/S430-2017, AISI S100-2017 and AISI S100-2018.
 - Loads shown in tables are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
 - Bending Moment (inches used for design) shall be Single and Two Span. $M^u = \frac{wL^2}{8}$ Three Span or More: $M^u = \frac{wL^2}{10}$
 - Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span condition.

STEEL CATALOG PAGE 71

Span	Single	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	68	63	58	53	48	43	38	33	28
	20	86	84	80	76	71	66	61	56	51
	18	106	95	83	71	57	46	37	30	24
	16	133	120	107	94	81	68	55	43	32
Double	22	104	104	99	94	89	84	79	74	69
	20	207	165	130	104	84	68	55	43	32
	18	303	238	176	138	111	90	74	62	52
	16	477	384	284	210	162	124	102	85	72
Triple	22	129	121	114	107	100	93	86	79	72
	20	162	142	124	106	89	73	59	48	39
	18	238	178	137	108	87	70	58	48	41
	16	327	246	189	149	119	97	80	66	56

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Total Slab Depth	Deck Type	Maximum Unshored Clear Span
15'-0"	15x6x20 ga	8'-0"
15'-0"	15x6x18 ga	7'-0"
15'-0"	15x6x16 ga	6'-0"
15'-0"	15x6x14 ga	5'-0"
15'-0"	15x6x12 ga	4'-0"
15'-0"	15x6x10 ga	3'-0"
15'-0"	15x6x8 ga	2'-0"
15'-0"	15x6x6 ga	1'-0"
15'-0"	15x6x4 ga	0'-0"
15'-0"	15x6x2 ga	0'-0"
15'-0"	15x6x0 ga	0'-0"
15'-0"	15x6x-2 ga	0'-0"
15'-0"	15x6x-4 ga	0'-0"
15'-0"	15x6x-6 ga	0'-0"
15'-0"	15x6x-8 ga	0'-0"
15'-0"	15x6x-10 ga	0'-0"
15'-0"	15x6x-12 ga	0'-0"
15'-0"	15x6x-14 ga	0'-0"
15'-0"	15x6x-16 ga	0'-0"
15'-0"	15x6x-18 ga	0'-0"
15'-0"	15x6x-20 ga	0'-0"
15'-0"	15x6x-22 ga	0'-0"
15'-0"	15x6x-24 ga	0'-0"
15'-0"	15x6x-26 ga	0'-0"
15'-0"	15x6x-28 ga	0'-0"
15'-0"	15x6x-30 ga	0'-0"
15'-0"	15x6x-32 ga	0'-0"
15'-0"	15x6x-34 ga	0'-0"
15'-0"	15x6x-36 ga	0'-0"
15'-0"	15x6x-38 ga	0'-0"
15'-0"	15x6x-40 ga	0'-0"
15'-0"	15x6x-42 ga	0'-0"
15'-0"	15x6x-44 ga	0'-0"
15'-0"	15x6x-46 ga	0'-0"
15'-0"	15x6x-48 ga	0'-0"
15'-0"	15x6x-50 ga	0'-0"
15'-0"	15x6x-52 ga	0'-0"
15'-0"	15x6x-54 ga	0'-0"
15'-0"	15x6x-56 ga	0'-0"
15'-0"	15x6x-58 ga	0'-0"
15'-0"	15x6x-60 ga	0'-0"
15'-0"	15x6x-62 ga	0'-0"
15'-0"	15x6x-64 ga	0'-0"
15'-0"	15x6x-66 ga	0'-0"
15'-0"	15x6x-68 ga	0'-0"
15'-0"	15x6x-70 ga	0'-0"
15'-0"	15x6x-72 ga	0'-0"
15'-0"	15x6x-74 ga	0'-0"
15'-0"	15x6x-76 ga	0'-0"
15'-0"	15x6x-78 ga	0'-0"
15'-0"	15x6x-80 ga	0'-0"
15'-0"	15x6x-82 ga	0'-0"
15'-0"	15x6x-84 ga	0'-0"
15'-0"	15x6x-86 ga	0'-0"
15'-0"	15x6x-88 ga	0'-0"
15'-0"	15x6x-90 ga	0'-0"
15'-0"	15x6x-92 ga	0'-0"
15'-0"	15x6x-94 ga	0'-0"
15'-0"	15x6x-96 ga	0'-0"
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